

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0898

Kettner 100 kV
June 19, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0898. The Interconnection Customer proposes a 104.3 MW solar photovoltaic facility interconnecting at the Kettner 100 kV substation in the Indiana area. The request was received on November 30, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 318 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,848,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ELLIOTT - GATES 100.0 kV	LA MARQUE - LAGRANGE 100.0 kV L	127.1	238	100.0
HALES BRANCH - ELLIOTT 100.0 kV	LA MRQUE - LAGRANGE 100.0 kV L	125.4	251	100.0
HALES BRANCH - KETTNER 100.0 kV Line	LA MARQUE - LAGRANGE 100.0 kV	121.8	286	100.0
KENDALLVILLE - FARMERSVILLE 100.0 kV	DEKALB - JONES CHAPEL 500.0 kV	120.3	264	7.9
SEGS 5 - HILLCREST 100.0 kV	DEKALB SUBSTATION - RICHLAND 500.0 kV	118.3	354	7.3
24TH & GARDENDALE - LANDSMAN CRK 230.0 kV	RICHLAND - DEKALB 500.0 kV	115.8	589	8.7
SIDNEY A MURRAY JR HYDROELECTRIC - 24TH & GARDENDALE SUBSTATION 230.0 kV	RICHLAND - DEKALB 500.0 kV	115.0	615	8.7
RICHLAND - DEKALB SUBSTATION 500.0 kV L	TREGO - SWAN VLY 500.0 kV	112.0	2,987	46.3
MINNEHAHA - NAPOLEON PEAKING STA 230.0 kV	TROY ENERGY LLC - RICHLAND 500.0 kV	111.2	977	12.0
CONSTANTINE HYDRO - FLOWERFIELD 100.0 kV	KETTNER - HALES BRANCH 100.0 kV	109.0	153	59.9
CONSTANTINE HYDRO - MEARS JCT 100.0 kV	HALES BRANCH - KETTNER 100.0 kV	108.6	271	100.0
LAGRANGE - STURGIS 100.0 kV	HALES BRANCH - KETTNER 100.0 kV	108.1	302	100.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MEARS JUNCTION - WEBSTER IC 100.0 kV	HALES BRANCH - KETTNER 100.0 kV	107.8	287	100.0
CUBA CITY - EGE 100.0 kV	JONES CHAPEL - DEKALB 500.0 kV	107.5	251	7.9
TREGO 230/1kV Autotransformer	TREGO - TROY ENERGY LLC 500.0 kV L	107.5	644	12.2
STUBEY ROAD - WEBSTER IC SUBSTATION 100.0 kV Line	HALES BRANCH - KETTNER SUBSTATION 100.0 kV	107.3	299	100.0
LA MARQUE - LAGRANGE 100.0 kV	HALES BRANCH - KETTNER 100.0 kV line	107.2	324	100.0
STURGIS - IDLE HOUR 100.0 kV	KETTNER - HALES BRANCH 100.0 kV	107.1	307	100.0
STUEY ROAD - IDLE HOUR 100.0 kV Line	HALES BRANCH SUBSTATION - KETTNER 100.0 kV	106.9	310	100.0
JAMES DE YOUNG - ST. LARENCE 230.0 kV Line	E METHUEN - LONETREE 500.0 kV	105.9	1,026	7.2
SWAN VALLEY - MUSKOGEE 500.0 kV line	RICHLAND - DEKALB 500.0 kV	105.7	2,943	23.9
LONTREE - SABANERA 500.0 kV Line	MUSKOGEE - MONTPELIER ELECTRIC GENERATING STA 500.0 kV Line	105.4	2,866	12.1
ADIRONDACK 500/230 kV TX	EAST METHUEN 500/230kV Autotransformer	105.2	1,026	5.3
HEBO - BARRETT 230.0 kV	WINTHROP - HEBO 230.0 kV	104.9	560	5.8
TREGO - HIGHGATE 230.0 kV L	TREGO SUBSTATION - TROY ENERGY LLC SUBSTATION 500.0 kV Line	104.4	511	10.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GATES - ELLIOTT 100.0 kV	Rebuild line with higher-capacity conductor	\$1,828,000	26% (\$473,000)
ELLIOT - HALES BRANCH SUBSTATION 100.0 kV	Rebuild line with higher-capacity conductor	\$8,530,000	26% (\$2,206,000)
HALES BRANCH - KETTNER 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,953,000	26% (\$1,022,000)
FARMERSVILLE - KENDALLVILLE 100.0 kV	Rebuild line with higher-capacity conductor	\$4,067,000	3% (\$117,000)
HILLCREST - SEGS 5 100.0 kV	Rebuild line with higher-capacity conductor	\$3,203,000	2% (\$50,000)
24TH & GARDENDALE - LANDSMAN CRK SUBSTATION 230.0 kV	Rebuild line with higher-capacity conductor	\$55,578,000	2% (\$1,140,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SIDNEY A MURRAY J HYDROELECTRIC - 24TH & GARDENDALE 230.0 kV	Rebuild line with higher-capacity conductor	\$20,949,000	2% (\$430,000)
RICHLAND - DEKALB SUBSTATION 500.0 kV L	Rebuild line with higher-capacity conductor	\$221,064,000	2% (\$3,910,000)
MINNEHAHA - NAPOLEON PEAKING STA 230.0 kV	Rebuild line with higher-capacity conductor	\$40,748,000	2% (\$706,000)
FLOWERFIELD SUBSTATION - CONSTANTINE HYDRO 100.0 kV	Reconductor line	\$3,065,000	26% (\$793,000)
CONSTANTINE HYDRO - MEARS JCT 100.0 kV	Reconductor line	\$2,321,000	26% (\$600,000)
LAGRANGE - STURGIS 100.0 kV	Reconductor line	\$4,752,000	26% (\$1,229,000)
WEBSTER IC - MEARS JUNCTION 100.0 kV	Reconductor line	\$962,000	26% (\$249,000)
EGE - CUBA CITY 100.0 kV Line	Reconductor line	\$1,805,000	3% (\$52,000)
TREGO 230/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$9,144,000	2% (\$157,000)
WEBSTER IC - STUBEY ROAD SUBSTATION 100.0 kV	Reconductor line	\$932,000	26% (\$241,000)
LA MARQUE - LAGRANGE 100.0 kV	Reconductor line	\$715,000	26% (\$185,000)
IDLE HOUR - STURGIS 100.0 kV L	Reconductor line	\$718,000	26% (\$186,000)
STUBEY RD - IDLE HOUR 100.0 kV	Reconductor line	\$1,784,000	26% (\$461,000)
ST. LAWRENCE - JAMES DE YOUNG SUBSTATION 230.0 kV Line	Reconductor line	\$25,874,000	1% (\$217,000)
SWAN VALLEY - MUSKOGEE 500.0 kV	Reconductor line	\$48,998,000	1% (\$541,000)
LONETREE - SABNERA 500.0 kV Line	Reconductor line	\$47,459,000	1% (\$300,000)
ADIRONDACK 500/230 kV TX	Replace transformer with higher-rated unit	\$34,562,000	1% (\$313,000)
HEBO - BARRETT 230.0 kV	Reconductor line	\$3,302,000	1% (\$32,000)
TREGO - HIGHGATE 230.0 kV L	Reconductor line	\$12,432,000	2% (\$238,000)
	Total allocated to project		\$15,848,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.